

PROJECT DOCUMENTATION

Online Quiz Application

Phase 1: Planning & Design | June 01–08, 2025

Academic Project Daily Log

Activity Summary — June 01 to June 08, 2025

Day	Date	Activity	Status
Day 1	June 01	Project Title Finalization	✓ Completed
Day 2	June 02	Requirement Gathering	✓ Completed
Day 3	June 03	Objective Definition	✓ Completed
Day 4	June 04	User & Module Identification	✓ Completed
Day 5	June 05	Use Case Diagram Preparation	✓ Completed
Day 6	June 06	Database Requirement Analysis	✓ Completed
Day 7	June 07	ER Diagram Design	✓ Completed
Day 8	June 08	Database Schema Creation	✓ Completed

Day 1 — June 01, 2025 | Project Title Finalization

Project Title	Online Quiz Application
Domain	Education / Web-Based Application
Technology Stack	Frontend: HTML/CSS/JS Backend: PHP / Node.js Database: MySQL

Project Overview

The Online Quiz Application is a web-based platform designed to allow administrators to create and manage quizzes, and students to attempt quizzes and view their results. The system supports multiple-choice questions, timed quizzes, role-based access, and automated score calculation.

Justification for Title

- **'Online'** — The application is fully web-based and accessible from any browser without installation.
- **'Quiz'** — Core functionality revolves around creating, managing, and attempting question sets.
- **'Application'** — A complete software system with user authentication, data management, and reporting.

Outcome: Title approved: Online Quiz Application. Scope confirmed and domain finalised.

Functional Requirements

- Admin can create, edit, and delete quizzes with a defined duration.
- Admin can add multiple-choice questions (4 options, 1 correct answer) to each quiz.
- Students can register, log in, and browse available quizzes.
- Students can attempt a quiz within the allotted time limit.
- System automatically calculates and stores the score upon quiz submission.
- Students can view their attempt history and detailed results.
- Admin can view all student attempts and performance reports.

Non-Functional Requirements

- Performance: Quiz pages and scoring should respond within 3 seconds.
- Security: Role-based login (Admin, Student) with hashed passwords.
- Usability: Clean, mobile-responsive UI accessible from any browser.
- Reliability: System should handle concurrent quiz sessions without data loss.
- Scalability: Should support up to 200 students and 50 quizzes simultaneously.

Outcome: Complete requirements document prepared covering all modules and constraints.

Primary Objectives

- To develop a web-based platform for creating and managing online quizzes.
- To provide students with a seamless interface for attempting timed quizzes.
- To automate score calculation and result generation upon quiz submission.
- To provide an administrative dashboard for complete quiz and user management.
- To maintain a digital record of all attempts and scores for reporting.

Secondary Objectives

- To improve academic assessment efficiency through automated grading.
- To provide students with instant feedback on their performance.
- To generate performance and participation reports for administrators.

Scope

In Scope: User authentication, quiz creation, question management, quiz attempts, score calculation, result viewing, admin reporting.

Out of Scope: Payment integration, live proctoring, essay-type or subjective questions.

Outcome: 5 primary and 3 secondary objectives documented. Scope boundaries clearly defined.

Day 4 — June 04, 2025 | User & Module Identification

User Roles

User Role	Description	Access Level
Admin	Creates and manages quizzes, questions, and views all reports	Full Access
Student	Registers, attempts quizzes, and views own results	Limited — Own Data Only

System Modules

Module	Key Features	Used By
User Management	Register, login, assign roles, manage profiles	Admin
Quiz Management	Create/edit/delete quizzes, set duration and status	Admin
Question Management	Add/edit/delete MCQ questions with correct answer	Admin
Attempt Module	Start quiz, answer questions, submit within timer	Student
Scoring Engine	Auto-calculate score and percentage on submission	System
Result Module	View score, correct/wrong answers, attempt history	Student, Admin
Report Module	Quiz-wise and student-wise performance analytics	Admin

Outcome: 2 user roles and 7 system modules identified and mapped.

Day 5 — June 05, 2025 | Use Case Diagram Preparation

Use Cases per Actor

Admin Use Cases:

- UC01 — Login to System
- UC02 — Manage Quizzes (Create / Edit / Delete / Publish)
- UC03 — Manage Questions (Add / Edit / Delete)
- UC04 — View All Students and Attempts
- UC05 — View Quiz Performance Reports
- UC06 — Manage User Accounts

Student Use Cases:

- UC07 — Register and Login to System
- UC08 — Browse Available Quizzes
- UC09 — Attempt a Quiz (Answer MCQs within Timer)
- UC10 — Submit Quiz and Receive Score
- UC11 — View Attempt History and Results

Use Case Diagram (Text Representation)

Actor	Use Cases — Online Quiz Application
Admin	Login Manage Quizzes Manage Questions View Reports Manage Users
Student	Register/Login Browse Quizzes Attempt Quiz Submit & Get Score View Results

Outcome: 11 use cases identified across 2 actors. Use case diagram prepared and reviewed.

Entities Identified

- Users — stores all user accounts (Admin, Student) with role-based access
- Quizzes — stores quiz details including title, description, and duration
- Questions — stores MCQ questions with all 4 options and the correct answer
- Attempts — records each student's quiz attempt with date and score
- Answers — stores the selected answer for each question per attempt
- Results — stores final score, total marks, and percentage per attempt

Data Requirements per Entity

Entity	Key Attributes	Relationships
Users	user_id, name, email, password, role	1 User (admin) creates many Quizzes; 1 User (student) makes many Attempts
Quizzes	quiz_id, title, description, duration, created_by	1 Quiz has many Questions; 1 Quiz has many Attempts
Questions	question_id, quiz_id, question_text, option_a–d, correct_answer	Many Questions belong to 1 Quiz
Attempts	attempt_id, user_id, quiz_id, attempt_date, score	1 Attempt by 1 User on 1 Quiz; 1 Attempt has many Answers
Answers	answer_id, attempt_id, question_id, selected_answer	Many Answers belong to 1 Attempt
Results	result_id, attempt_id, total_marks, obtained_marks, percentage	1 Result per 1 Attempt

Outcome: 6 entities identified with all key attributes and inter-entity relationships mapped.

Day 7 — June 07, 2025 | ER Diagram Design

Entities and Attributes

USERS		QUIZZES	
user_id (PK)	Primary Key	quiz_id (PK)	Primary Key
name	Attribute	title	Attribute
email	Attribute	description	Attribute
password (hashed)	Attribute	duration	Attribute
role (admin/student)	Attribute	created_by (FK → Users)	Foreign Key

QUESTIONS		ATTEMPTS	
question_id (PK)	Primary Key	attempt_id (PK)	Primary Key
quiz_id (FK → Quizzes)	Foreign Key	user_id (FK → Users)	Foreign Key
question_text	Attribute	quiz_id (FK → Quizzes)	Foreign Key
option_a / b / c / d	Attribute	attempt_date	Attribute
correct_answer	Attribute	score	Attribute

ANSWERS		RESULTS	
answer_id (PK)	Primary Key	result_id (PK)	Primary Key
attempt_id (FK → Attempts)	Foreign Key	attempt_id (FK → Attempts)	Foreign Key
question_id (FK → Questions)	Foreign Key	total_marks	Attribute
selected_answer	Attribute	obtained_marks	Attribute
		percentage	Attribute

Relationships Summary

From Entity	Relationship	To Entity	Type
Users	creates	Quizzes	One-to-Many
Users	makes	Attempts	One-to-Many
Quizzes	contains	Questions	One-to-Many
Quizzes	undergoes	Attempts	One-to-Many
Attempts	includes	Answers	One-to-Many
Attempts	generates	Results	One-to-One
Questions	referenced in	Answers	One-to-Many

Outcome: ER diagram designed with 6 entities, all primary/foreign keys defined, and 7 relationships mapped.

Day 8 — June 08, 2025 | Database Schema Creation

The following SQL tables are being created based on the ER diagram finalised on Day 7. The schema is implemented in MySQL for the Online Quiz Application.

Users Table

```
CREATE TABLE users (  
  user_id INT PRIMARY KEY AUTO_INCREMENT,  
  name VARCHAR(100) NOT NULL,  
  email VARCHAR(100) UNIQUE NOT NULL,  
  password VARCHAR(255) NOT NULL,  
  role ENUM('admin', 'student') DEFAULT 'student'  
);
```

Quizzes Table

```
CREATE TABLE quizzes (  
  quiz_id INT PRIMARY KEY AUTO_INCREMENT,  
  title VARCHAR(150) NOT NULL,  
  description TEXT,  
  duration INT NOT NULL,  
  created_by INT,  
  FOREIGN KEY (created_by) REFERENCES users(user_id)  
);
```

Questions Table

```
CREATE TABLE questions (  
  question_id INT PRIMARY KEY AUTO_INCREMENT,  
  quiz_id INT NOT NULL,  
  question_text TEXT NOT NULL,  
  option_a VARCHAR(255) NOT NULL,  
  option_b VARCHAR(255) NOT NULL,  
  option_c VARCHAR(255) NOT NULL,  
  option_d VARCHAR(255) NOT NULL,  
  correct_answer CHAR(1) NOT NULL,  
  FOREIGN KEY (quiz_id) REFERENCES quizzes(quiz_id)  
);
```

Attempts Table

```
CREATE TABLE attempts (  
  attempt_id INT PRIMARY KEY AUTO_INCREMENT,  
  user_id INT NOT NULL,  
  quiz_id INT NOT NULL,  
  attempt_date DATETIME DEFAULT CURRENT_TIMESTAMP,
```

```
score DECIMAL(5,2) DEFAULT 0,  
FOREIGN KEY (user_id) REFERENCES users(user_id),  
FOREIGN KEY (quiz_id) REFERENCES quizzes(quiz_id)  
);
```

Answers Table

```
CREATE TABLE answers (  
  answer_id INT PRIMARY KEY AUTO_INCREMENT,  
  attempt_id INT NOT NULL,  
  question_id INT NOT NULL,  
  selected_answer CHAR(1),  
  FOREIGN KEY (attempt_id) REFERENCES attempts(attempt_id),  
  FOREIGN KEY (question_id) REFERENCES questions(question_id)  
);
```

Results Table

```
CREATE TABLE results (  
  result_id INT PRIMARY KEY AUTO_INCREMENT,  
  attempt_id INT UNIQUE NOT NULL,  
  total_marks INT NOT NULL,  
  obtained_marks INT NOT NULL,  
  percentage DECIMAL(5,2),  
  FOREIGN KEY (attempt_id) REFERENCES attempts(attempt_id)  
);
```

Status

Database schema creation is **completed** as of June 08, 2025. All 6 tables have been created and verified in MySQL Workbench.